

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
pd.options.display.max_columns=99
```

```
df=pd.read_csv("/content/Mall_Customers.csv")
```

```
df.head()
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	
0	1	Male	19	15	39	
1	2	Male	21	15	81	
2	3	Female	20	16	6	
3	4	Female	23	16	77	
4	5	Female	31	17	40	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.tail()
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)	
195	196	Female	35	120	79	
196	197	Female	45	126	28	
197	198	Male	32	126	74	
198	199	Male	32	137	18	
199	200	Male	30	137	83	

```
df.shape
```

```
(200, 5)
```

```
df.describe()
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            200 non-null   int64
1   Gender                200 non-null   object
2   Age                  200 non-null   int64
3   Annual Income (k$)    200 non-null   int64
4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

```
plt.style.use("fivethirtyeight")
```

```
x_question=df[['Age','Annual Income (k$)','Spending Score (1-100)']]  
q_mean=x_question.mean(axis=0)  
total_mean=q_mean.mean
```

```
q_mean=q_mean.to_frame('mean')  
q_mean.reset_index(level=0,inplace=True)  
q_mean.head()
```

	index	mean
0	Age	38.85
1	Annual Income (k\$)	60.56
2	Spending Score (1-100)	50.20

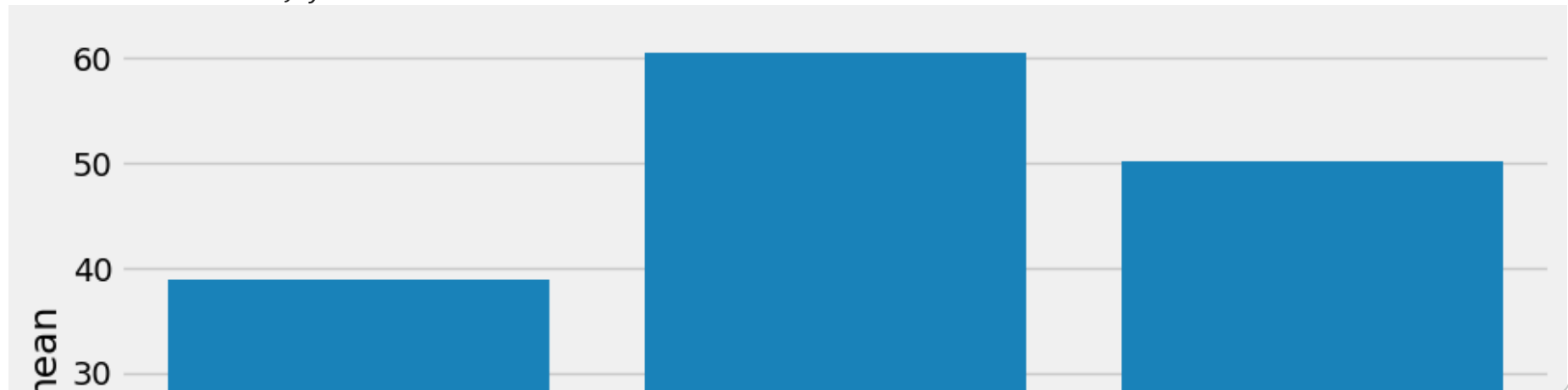


Next steps:

[Generate code with q_mean](#)[New interactive sheet](#)

```
plt.figure(figsize=(10,5))  
sns.barplot(x='index',y='mean',data=q_mean)
```

<Axes: xlabel='index', ylabel='mean'>



```
corr=df.corr(numeric_only=True)
```

```
plt.figure(figsize=(10,5))
sns.heatmap(corr,annot=True,cmap="coolwarm")
```

<Axes: >



```
x=df[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']]
```

```
from sklearn.decomposition import PCA
pca=PCA(n_components=2,random_state=42)
x_principalComponents=pca.fit_transform(x)
```

```
x_principalComponents
np.cumsum(pca.explained_variance_ratio_)[1]
```

```
np.float64(0.8922373735506912)
```

```

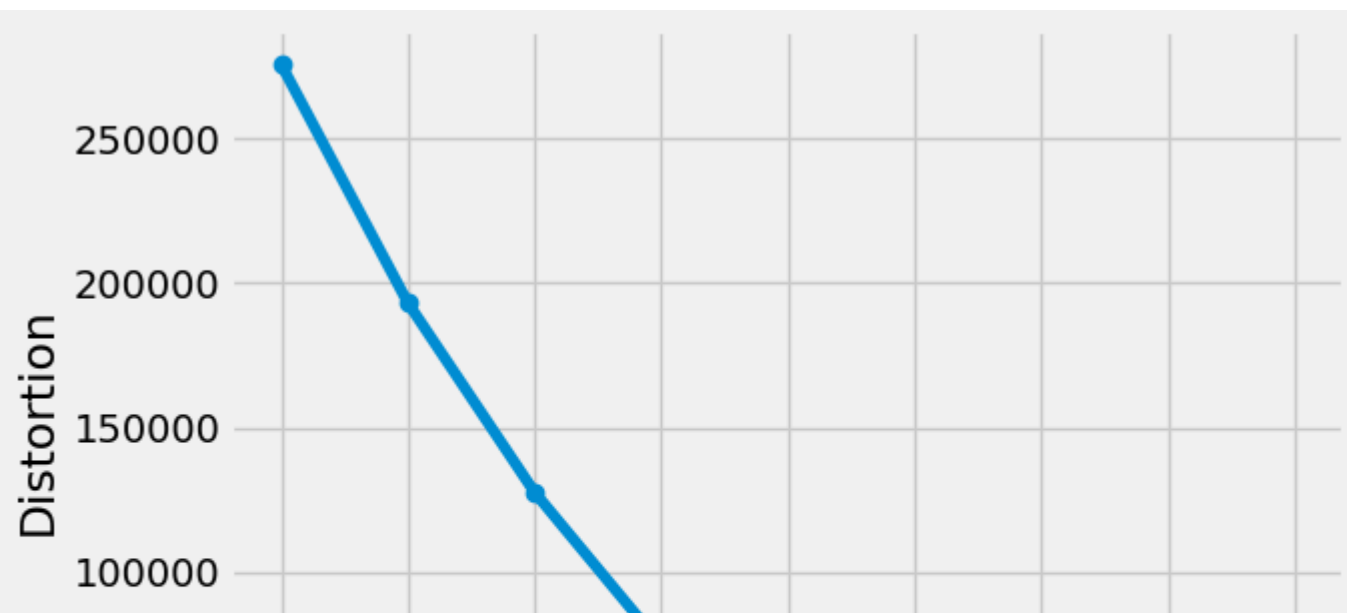
from sklearn.cluster import KMeans
distortion=[]
cluster_range=range(1,10)
for i in cluster_range:
    km=KMeans(n_clusters=i,random_state=42)
    km.fit(x_principalComponents)
    distortion.append(km.inertia_)

```

```

plt.plot(cluster_range,distortion,marker='o')
plt.xlabel('Number of Clusters')
plt.ylabel('Distortion')
plt.show()

```



```

model=KMeans(n_clusters=5,random_state=42)
model.fit(x_principalComponents)
y=model.predict(x_principalComponents)

```

```

plt.scatter(x_principalComponents[:,0],x_principalComponent1)
plt.xlabel('Principal Component 1')

```

```
plt.ylabel('Principal Component 2')  
plt.show()
```

